Practical 10

AIM : Demo of OpenAI/TensorFlow Tools

- Explore and experiment with OpenAI or TensorFlow tools and libraries.
- Perform a demonstration or mini-project showcasing the capabilities of the tools.
- Discuss and present the findings and potential applications.

CODE :

Intents.json

```
{
  "intents": [
    {
      "tag": "greeting",
      "patterns": ["hello", "hi", "hey", "good morning", "good evening", "how are
you", "hello chatbot"],
      "responses": ["Hello! How can I help you?", "Hi! Ask me an AI or ML question.",
"Welcome! Let's explore Artificial Intelligence."]
    },
    {
      "tag": "goodbye",
      "patterns": ["bye", "goodbye", "see you", "see you later"],
      "responses": ["Goodbye! Have a nice day.", "See you again!", "Thanks for
chatting with me."]
    },
    {
      "tag": "thanks",
      "patterns": ["thanks", "thank you", "thanks a lot", "thank you so much"],
      "responses": ["You're welcome!", "Happy to help!", "Glad I could help."]
    },
    {
      "tag": "help",
      "patterns": ["help", "help me", "what can I ask", "show available topics", "what
topics do you know"],
      "responses": ["Ask me about AI, Machine Learning, Deep Learning, Python,
TensorFlow, neural networks, NLP, computer vision, algorithms, training, testing, and
model evaluation."]
    },
    {
      "tag": "ai",
      "patterns": ["what is artificial intelligence", "define AI", "explain artificial
intelligence", "what is AI", "tell me about AI", "how does AI work"],
      "responses": ["Artificial Intelligence is a branch of computer science that
enables machines to perform tasks requiring human-like intelligence.", "AI systems can
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perform tasks such as recognizing images, understanding language, making predictions,
and solving problems."]
    },
    {
      "tag": "types_of_ai",
      "patterns": ["what are the types of AI", "types of artificial intelligence",
"explain narrow AI", "what is general AI", "classify AI"],
      "responses": ["AI is commonly categorized as narrow AI, which performs specific
tasks, and hypothetical general AI, which would have broad human-like capabilities.
Another classification uses reactive machines, limited-memory systems, and other
theoretical categories."]
    },
    {
      "tag": "applications",
      "patterns": ["applications of AI", "uses of artificial intelligence", "where is
AI used", "real world applications of AI", "how is AI useful"],
      "responses": ["AI is used in healthcare, agriculture, education, banking,
robotics, transportation, cybersecurity, and customer support."]
    },
    {
      "tag": "advantages_ai",
      "patterns": ["advantages of AI", "benefits of artificial intelligence", "why is
AI useful", "what are the benefits of AI"],
      "responses": ["AI can automate repetitive tasks, analyze large datasets, support
decision-making, improve productivity, and provide services around the clock."]
    },
    {
      "tag": "limitations_ai",
      "patterns": ["disadvantages of AI", "limitations of AI", "problems with
artificial intelligence", "risks of AI"],
      "responses": ["AI may produce incorrect results, reflect bias in training data,
raise privacy concerns, require substantial computing resources, and automate some
jobs."]
    },
    {
      "tag": "ml",
      "patterns": ["what is machine learning", "define machine learning", "explain
ML", "what is ML", "tell me about machine learning"],
      "responses": ["Machine Learning is a branch of AI in which computers learn
patterns from data to make predictions or decisions."]
    },
    {
      "tag": "types_of_ml",
      "patterns": ["types of machine learning", "what are the types of ML", "explain
supervised learning", "explain unsupervised learning", "what is reinforcement
learning"],
      "responses": ["The main types of Machine Learning are supervised learning,
unsupervised learning, and reinforcement learning. They learn from labeled examples,
unlabeled data, and rewards or penalties respectively."]
    },
    {
      "tag": "supervised_learning",
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      "patterns": ["what is supervised learning", "define supervised learning",
"examples of supervised learning", "how does supervised learning work"],
      "responses": ["Supervised learning trains a model using labeled examples. Common
tasks include classification and regression. Examples include spam detection and
house-price prediction."]
    },
    {
      "tag": "unsupervised_learning",
      "patterns": ["what is unsupervised learning", "define unsupervised learning",
"examples of unsupervised learning", "what is clustering"],
      "responses": ["Unsupervised learning finds patterns in unlabeled data.
Clustering customers into groups is a common example."]
    },
    {
      "tag": "reinforcement_learning",
      "patterns": ["what is reinforcement learning", "define reinforcement learning",
"how does reinforcement learning work", "examples of reinforcement learning"],
      "responses": ["Reinforcement learning trains an agent to choose actions by
receiving rewards or penalties from its environment. It is used in robotics, games,
and control systems."]
    },
    {
      "tag": "classification",
      "patterns": ["what is classification in ML", "explain classification",
"classification algorithm", "give an example of classification"],
      "responses": ["Classification predicts a category or class. Examples include
classifying an email as spam or not spam and identifying an image as a cat or dog."]
    },
    {
      "tag": "regression",
      "patterns": ["what is regression in ML", "explain regression", "regression
algorithm", "example of regression"],
      "responses": ["Regression predicts a numerical value. For example, linear
regression can estimate a house price from its features."]
    },
    {
      "tag": "clustering",
      "patterns": ["what is clustering in machine learning", "explain clustering",
"clustering algorithms", "what is K means"],
      "responses": ["Clustering groups similar data points without requiring
predefined labels. K-means is a widely used clustering algorithm."]
    },
    {
      "tag": "algorithms",
      "patterns": ["machine learning algorithms", "list ML algorithms", "which
algorithms are used in ML", "name some machine learning algorithms"],
      "responses": ["Common ML algorithms include Linear Regression, Logistic
Regression, Decision Trees, Random Forests, K-Nearest Neighbors, Support Vector
Machines, K-means, and Naive Bayes."]
    },
    {
      "tag": "decision_tree",
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      "patterns": ["what is a decision tree", "explain decision tree algorithm", "how
does a decision tree work", "uses of decision trees"],
      "responses": ["A Decision Tree makes predictions using a tree-like structure of
feature-based decisions. It can be used for both classification and regression."]
    },
    {
      "tag": "random_forest",
      "patterns": ["what is random forest", "explain random forest algorithm", "how
does random forest work"],
      "responses": ["Random Forest combines predictions from multiple decision trees.
It often improves generalization compared with a single tree."]
    },
    {
      "tag": "knn",
      "patterns": ["what is KNN", "explain K nearest neighbors", "how does KNN work",
"KNN algorithm"],
      "responses": ["K-Nearest Neighbors predicts a class or value using nearby
training examples. The parameter K specifies how many neighbors are considered."]
    },
    {
      "tag": "svm",
      "patterns": ["what is SVM", "explain support vector machine", "how does SVM
work", "support vector machine algorithm"],
      "responses": ["A Support Vector Machine finds a decision boundary that separates
classes. Kernel functions can help it model nonlinear boundaries."]
    },
    {
      "tag": "naive_bayes",
      "patterns": ["what is naive bayes", "explain naive bayes algorithm", "how does
naive bayes work", "naive bayes classifier"],
      "responses": ["Naive Bayes is a probabilistic classifier based on Bayes'
theorem. It assumes features are conditionally independent given the class."]
    },
    {
      "tag": "linear_regression",
      "patterns": ["what is linear regression", "explain linear regression", "linear
regression example", "how does linear regression work"],
      "responses": ["Linear Regression models the relationship between input features
and a numerical target using a linear equation."]
    },
    {
      "tag": "logistic_regression",
      "patterns": ["what is logistic regression", "explain logistic regression",
"logistic regression algorithm"],
      "responses": ["Logistic Regression estimates class probabilities and is commonly
used for binary classification."]
    },
    {
      "tag": "overfitting",
      "patterns": ["what is overfitting", "explain overfitting in ML", "how to prevent
overfitting", "why does a model overfit"],
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      "responses": ["Overfitting occurs when a model learns training data too closely
and performs poorly on unseen data. Regularization, dropout, more training data, and
cross-validation can help."]
    },
    {
      "tag": "underfitting",
      "patterns": ["what is underfitting", "explain underfitting", "how to prevent
underfitting"],
      "responses": ["Underfitting occurs when a model is too simple or insufficiently
trained to learn important patterns. A more suitable model or better training may
help."]
    },
    {
      "tag": "training_testing",
      "patterns": ["what is training data", "what is testing data", "difference
between training and testing data", "why split a dataset"],
      "responses": ["Training data teaches the model. Testing data evaluates
performance on examples not used for fitting. A separate validation set can be used to
tune model settings."]
    },
    {
      "tag": "accuracy",
      "patterns": ["what is model accuracy", "define accuracy in machine learning",
"how to calculate accuracy", "what is classification accuracy"],
      "responses": ["Accuracy is the proportion of predictions that are correct. It
can be misleading on imbalanced datasets, so precision, recall, and F1-score may also
be useful."]
    },
    {
      "tag": "precision_recall",
      "patterns": ["what is precision and recall", "explain precision", "explain
recall", "difference between precision and recall"],
      "responses": ["Precision measures how many predicted positives are truly
positive. Recall measures how many actual positives the model identifies."]
    },
    {
      "tag": "deep_learning",
      "patterns": ["what is deep learning", "define deep learning", "difference
between ML and deep learning", "explain deep learning"],
      "responses": ["Deep Learning is a branch of Machine Learning that uses neural
networks with multiple layers to learn complex patterns."]
    },
    {
      "tag": "neural_network",
      "patterns": ["what is a neural network", "explain neural networks", "what is an
artificial neural network", "how does a neural network work"],
      "responses": ["A neural network contains interconnected computational units
arranged in layers. It learns patterns by adjusting weights during training."]
    },
    {
      "tag": "activation_functions",
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      "patterns": ["what is an activation function", "explain ReLU", "what is sigmoid
activation", "why use activation functions"],
      "responses": ["Activation functions introduce nonlinear behavior into neural
networks. Common examples include ReLU, sigmoid, and tanh."]
    },
    {
      "tag": "epochs",
      "patterns": ["what is an epoch", "define epoch in deep learning", "what is batch
size", "what is an iteration in ML"],
      "responses": ["An epoch is one complete pass through the training dataset. Batch
size is the number of examples processed together in one training step."]
    },
    {
      "tag": "loss_function",
      "patterns": ["what is a loss function", "explain loss in machine learning", "why
is loss calculated", "what is categorical cross entropy"],
      "responses": ["A loss function measures the difference between a model's
predictions and target values. Training aims to minimize this loss."]
    },
    {
      "tag": "optimizer",
      "patterns": ["what is an optimizer", "explain Adam optimizer", "what is gradient
descent", "why use an optimizer"],
      "responses": ["An optimizer updates model parameters to reduce the loss. Common
choices include Gradient Descent, SGD, and Adam."]
    },
    {
      "tag": "tensorflow",
      "patterns": ["what is tensorflow", "explain tensorflow", "why use tensorflow",
"what is tensorflow used for"],
      "responses": ["TensorFlow is an open-source machine learning framework used to
build, train, and deploy machine learning models."]
    },
    {
      "tag": "keras",
      "patterns": ["what is keras", "explain keras", "difference between tensorflow
and keras", "what is tf keras"],
      "responses": ["Keras provides a high-level interface for building and training
neural networks. TensorFlow includes tf.keras for this purpose."]
    },
    {
      "tag": "python",
      "patterns": ["what is python", "tell me about python", "why use python",
"advantages of python"],
      "responses": ["Python is a high-level programming language widely used in AI,
machine learning, web development, automation, and data science."]
    },
    {
      "tag": "numpy",
      "patterns": ["what is numpy", "explain numpy", "why use numpy in ML"],
      "responses": ["NumPy is a Python library for numerical computing. It provides
arrays and mathematical operations useful for machine learning."]
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    },
    {
      "tag": "nlp",
      "patterns": ["what is NLP", "define NLP", "what is natural language processing",
"explain NLP", "how do computers process language"],
      "responses": ["Natural Language Processing enables computers to process and
analyze human language. Applications include chatbots, translation, and sentiment
analysis."]
    },
    {
      "tag": "tokenization",
      "patterns": ["what is tokenization", "explain tokenization in NLP", "what are
tokens", "how does tokenization work"],
      "responses": ["Tokenization splits text into smaller units such as words,
subwords, or characters for further processing."]
    },
    {
      "tag": "bag_of_words",
      "patterns": ["what is bag of words", "explain bag of words", "how does bag of
words work"],
      "responses": ["Bag of Words represents text using word occurrence or frequency
features while ignoring most word-order information."]
    },
    {
      "tag": "computer_vision",
      "patterns": ["what is computer vision", "explain computer vision", "applications
of computer vision", "how does computer vision work"],
      "responses": ["Computer Vision enables computers to interpret images and videos.
Applications include object detection, facial recognition, and medical image
analysis."]
    },
    {
      "tag": "cnn",
      "patterns": ["what is CNN in deep learning", "explain convolutional neural
network", "what are CNNs used for"],
      "responses": ["A Convolutional Neural Network is a neural network commonly used
for image processing because it learns spatial features such as edges and shapes."]
    },
    {
      "tag": "generative_ai",
      "patterns": ["what is generative AI", "explain generative artificial
intelligence", "how does generative AI work", "examples of generative AI"],
      "responses": ["Generative AI models create new content such as text, images,
audio, or code based on patterns learned from data."]
    },
    {
      "tag": "chatbot",
      "patterns": ["what is a chatbot", "define chatbot", "how does a chatbot work",
"what can you do", "who are you"],
      "responses": ["I am a TensorFlow chatbot that predicts the intent of your
message and selects a predefined response from my dataset."]
    },
```

```
    {
      "tag": "applications_ml",
      "patterns": ["applications of machine learning", "uses of ML", "where is machine
learning used", "real world ML examples"],
      "responses": ["Machine Learning is used for spam filtering, fraud detection,
recommendations, demand forecasting, medical analysis, and image classification."]
    },
    {
      "tag": "overfitting_vs_underfitting",
      "patterns": ["difference between overfitting and underfitting", "compare
overfitting and underfitting", "overfitting versus underfitting"],
      "responses": ["Overfitting means a model fits training data too closely and
generalizes poorly. Underfitting means it fails to learn important patterns even from
the training data."]
    }
  ]
}
```

Chatbot.py

```
# AI Chatbot Using TensorFlow and NLP

import json
import random
import re

import numpy as np
import tensorflow as tf

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Input, Dense

# 1. Set random seeds for reproducibility

random.seed(42)
np.random.seed(42)
tf.random.set_seed(42)

# 2. Load the intent dataset

with open("intents.json", "r", encoding="utf-8") as file:
    data = json.load(file)

# 3. Prepare words, classes, and training examples

words = []
classes = []
documents = []

for intent in data["intents"]:
    tag = intent["tag"]
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    if tag not in classes:
        classes.append(tag)

    for pattern in intent["patterns"]:
        tokens = re.findall(r"\b\w+\b", pattern.lower())

        words.extend(tokens)
        documents.append((tokens, tag))

words = sorted(set(words))
classes = sorted(set(classes))

print("Total training examples:", len(documents))
print("Total unique words:", len(words))
print("Total intent classes:", len(classes))

# 4. Convert text into a bag-of-words vector

def bag_of_words(tokens):
    token_set = set(tokens)

    return np.array(
        [
            1.0 if word in token_set else 0.0
            for word in words
        ],
        dtype=np.float32
    )

# 5. Create training data

X = []
y = []

for tokens, tag in documents:
    X.append(bag_of_words(tokens))
    y.append(classes.index(tag))

X = np.array(X, dtype=np.float32)
y = np.array(y, dtype=np.int32)

# 6. Build the TensorFlow neural network

model = Sequential([
    Input(shape=(len(words),)),
    Dense(32, activation="relu"),
    Dense(16, activation="relu"),
    Dense(len(classes), activation="softmax")
])

# 7. Compile the model
```


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model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

# 8. Train the model

print("\nTraining the chatbot model...")

model.fit(
    X,
    y,
    epochs=250,
    batch_size=8,
    verbose=0
)

print("Model training completed successfully!")

# 9. Define the response function
def get_response(message):
    # Convert the user's message into tokens
    tokens = re.findall(r"\b\w+\b", message.lower())
    if not tokens:
        return "Please enter a question."
    # Check whether any input word is known to the model
    known_tokens = [word for word in tokens if word in words]
    if not known_tokens:
        return (
            "Sorry, I don't understand that question. "
            "Type 'help' to see the available topics."
        )
    # Predict the intent probabilities
    input_vector = bag_of_words(tokens).reshape(1, -1)
    probabilities = model.predict(input_vector, verbose=0)[0]
    predicted_index = int(np.argmax(probabilities))
    confidence = float(probabilities[predicted_index])
    # Reject low-confidence predictions
    if confidence < 0.65:
        return (
            "I'm not sure I understand. "
            "Please rephrase your question or type 'help'."
        )
    predicted_tag = classes[predicted_index]
    # Find a suitable response for the predicted intent
    for intent in data["intents"]:
        if intent["tag"] == predicted_tag:
            return random.choice(intent["responses"])
    return "Sorry, I couldn't find a suitable response."
# 10. Run the chatbot
```

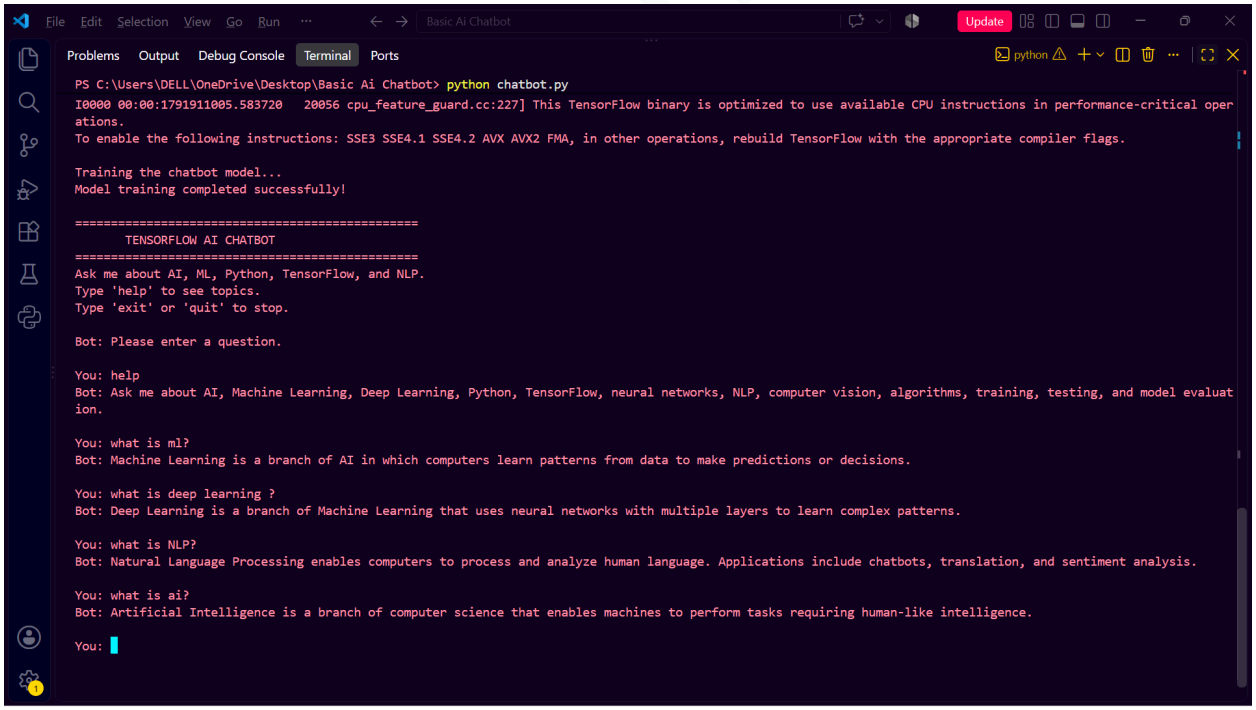
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```
def main():
    print("\n" + "=" * 48)
    print("       TENSORFLOW AI CHATBOT")
    print("=" * 48)
    print("Ask me about AI, ML, Python, TensorFlow, and NLP.")
    print("Type 'help' to see topics.")
    print("Type 'exit' or 'quit' to stop.\n")
    while True:
        try:
            message = input("You: ").strip()
        except (KeyboardInterrupt, EOFError):
            print("\nBot: Goodbye!")
            break
        if message.lower() in ("exit", "quit"):
            print("Bot: Goodbye! Have a nice day.")
            break
        response = get_response(message)
        print("Bot:", response, "\n")

if __name__ == "__main__":
    main()
```

OUTPUT :



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